



Terms of Reference (ToR) – Long -Term Agreement (LTA) Consultancy services for Vertical Electrical Sounding (VES) and Electrical Resistivity Tomography (ERT) study in WaterAid Nepal working areas

Introduction

WaterAid is an International Non-profit organization with one goal: to change the world through water. Along with decent toilets and good hygiene, a reliable supply of clean water is essential for health, dignity and a life full of opportunity. Childbirth is safer, and children get the chance to grow up healthy and strong. Families thrive, and whole communities are better prepared for our changing climate and whatever the future holds. We started in 1981, when UK water companies created a single organization dedicated to tackling the global water crisis. Today, we work alongside communities worldwide, setting up entire systems that deliver clean water, decent toilets and good hygiene for millions. And we get the people, policies and money in place to keep these systems working – and the water flowing for good. So that people have safe, reliable services that last. Our work with communities, partners, allies, and supporters has already enabled millions of people to shape their futures with dignity and confidence.

Our vision is ‘A world where everyone, everywhere has sustainable and safe water, sanitation and hygiene’.

WaterAid has been working in Nepal since 1987, covering all seven provinces. We team up with local partners, technical experts, governments, and communities to support the delivery of integrated and inclusive WASH services in hard-to-reach communities, helping to transform lives.

Background

WaterAid Nepal (WAN) works to improve access to safe, reliable, and sustainable water, sanitation, and hygiene (WASH) services, particularly for marginalized and underserved communities. In Nepal's evolving context, where many rural settlements are transitioning into emerging municipalities and urban centres, the demand for reliable, efficient, and sustainable water infrastructure is steadily increasing, particularly in the Terai region, including Madhesh Province.

One of the key objectives of WaterAid Nepal's programmes is to strengthen the operational efficiency, planning capacity, and long-term sustainability of water utilities and Water Supply and Sanitation User Committees (WSSUCs). Groundwater is a critical natural resource for drinking water supply, irrigation, and industrial use across Nepal, particularly in the Terai region where many urban and emerging municipal water supply systems depend on groundwater abstraction through production boreholes. Aquifers are recharged primarily through rainfall infiltration and river systems and represent one of the most reliable sources of year-round water supply. The productivity and sustainability of groundwater resources depend on local geological and hydrogeological conditions, including aquifer thickness, lithological composition, permeability, and recharge characteristics.

In many municipalities and service areas supported by WaterAid Nepal and Nepal Water Supply Corporation (NWSC), limited hydrogeological information is available readily to support the identification of suitable borehole locations, aquifer characterization, groundwater monitoring, and water source development planning. Geophysical investigation techniques such as Vertical Electrical Sounding (VES) and Electrical Resistivity Tomography (ERT) provide reliable and cost-effective approaches for understanding subsurface conditions, delineating aquifer formations, estimating groundwater potential, and informing borehole drilling decisions.

Currently, Nepal Water Supply Corporation (NWSC) operates branch offices in nine towns of Madhesh Province: Rajbiraj, Lahan, Janakpur, Jaleshwar, Gaushala, Malangwa, Gaur, Kalaiya, and Birgunj.

To strengthen groundwater resource planning and support sustainable water supply services, WaterAid Nepal intends to establish a Long-Term Agreement (LTA) for a period of three years with qualified consulting agencies to undertake hydrogeological investigations, VES surveys, ERT surveys, groundwater assessments, borehole siting studies, and related technical advisory services across WaterAid Nepal's programme areas.

The LTA will serve as a framework arrangement under which WAN may issue assignment-specific Terms of Reference (ToR) and work orders during the agreement period, based on programmatic priorities and emerging needs.

Purpose of the Long-Term Agreement

The purpose of this LTA is to establish a roster-based contractual arrangement with one or more qualified consulting agencies to provide hydrogeological investigation services, including Vertical Electrical Sounding (VES), Electrical Resistivity Tomography (ERT), groundwater assessments, borehole siting investigations, aquifer characterization, and technical advisory services on an as-needed basis during the agreement period. The LTA shall provide a mechanism for WaterAid Nepal to issue assignment-specific work orders without undertaking separate procurement processes for each assignment. However, this LTA does not guarantee any minimum quantity or value of work during the contract period.

Objective of the assignment

Overall Objective:

The overall objective of this assignment is to support sustainable groundwater resource development and evidence-based water supply planning through hydrogeological investigations, groundwater assessments, aquifer characterization, borehole siting studies, and geophysical surveys in WaterAid Nepal programme areas.

Specific objectives

The specific objectives of the assignment specific works will be to:

- I. Conduct Vertical Electrical Sounding (VES) surveys and Electrical Resistivity Tomography (ERT) surveys to assess groundwater potential in selected project areas.
- II. Identify hydrogeological formations, aquifer characteristics, and subsurface lithological conditions.
- III. Determine suitable locations, depths, and drilling methods for production boreholes.
- IV. Support groundwater source development, monitoring, and protection planning.
- V. Generate technical evidence to support NWSC, municipalities, WSSUCs, and WaterAid Nepal in groundwater resource management and water infrastructure investments.

Methodology

The consulting agency shall deploy a multidisciplinary team comprising, at a minimum, a **Team Leader/Senior Hydrogeologist, Geophysicist, Hydrogeologist, Field Survey Engineers/Technicians, and Data Processing & GIS Specialist** to undertake the assignment. The consulting agency will work closely with WaterAid Nepal, Nepal Water Supply Corporation (NWSC), Water Supply and Sanitation User Committees (WSSUCs), municipalities, and other relevant stakeholders throughout the assignment.

The consulting agency shall be responsible for coordinating with WaterAid Nepal, implementing partners, government counterparts, municipalities, and other concerned stakeholders, as required, to ensure the effective planning, implementation, quality assurance, and delivery of the assignment outputs. The assignment shall be carried out in accordance with applicable national standards, professional best practices, and WaterAid Nepal's programme requirements.

The consulting agency shall adopt a systematic workflow for VES and ERT surveying, as outlined below.

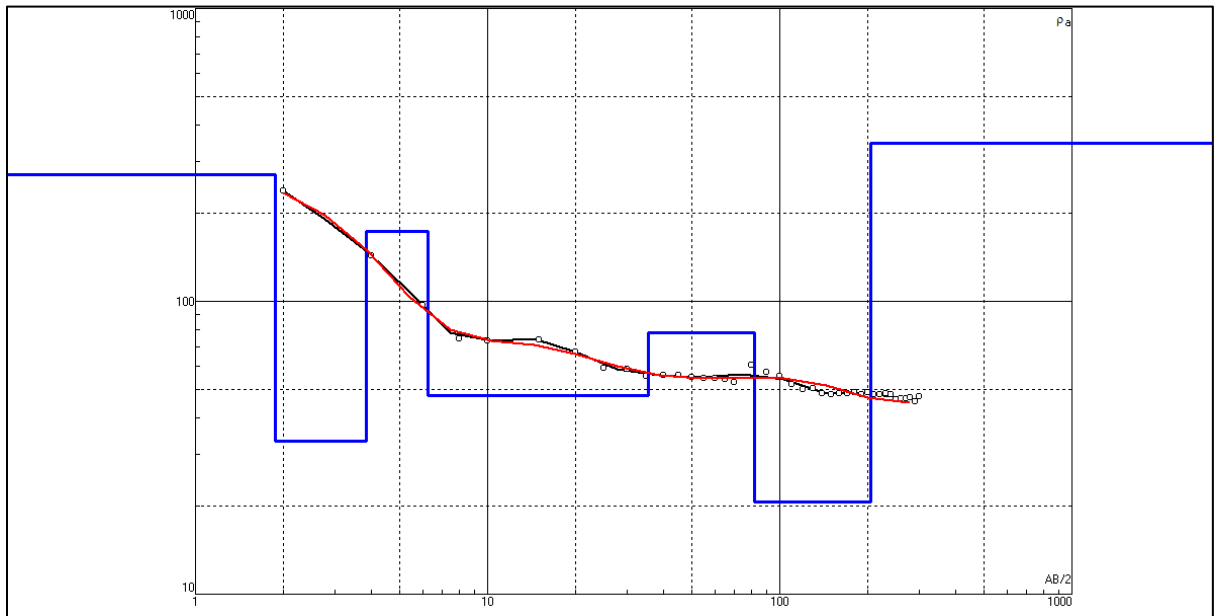
Part -A: Desk Review and Geological Investigation

- Review of existing hydrogeological and geological reports.
- Collection of drilling logs, lithological records, borehole completion reports and pumping test data.
- Review of groundwater abstraction and recharge information.
- Assessment of regional geology and aquifer systems.
- Preparation of preliminary conceptual hydrogeological model.

Part -B: Vertical Electrical Sounding (VES)

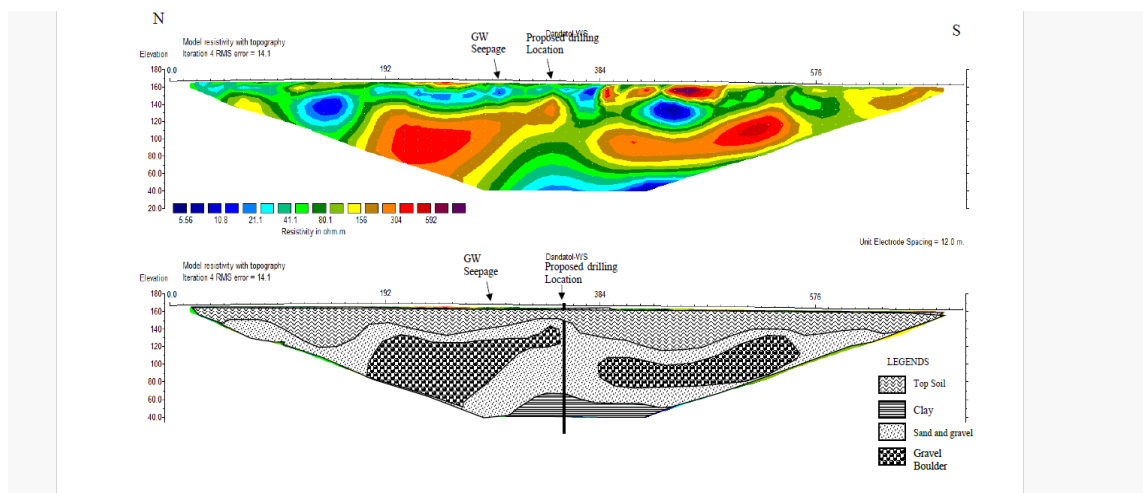
- VES surveys using Schlumberger electrode configuration.
- Electrode spacing based on required investigation depth.
- Maximum current electrode spacing of not less than 800 m or according to site requirements.
- Generation and interpretation of VES curves using appropriate software.
- Estimation of layer thickness, resistivity values and subsurface lithology.

- Identification of potential aquifer zones and recommended drilling depths.



Part -C: Electrical Resistivity Tomography (ERT)

- ERT survey profiles designed according to site conditions and hydrogeological objectives.
- Alignment of profiles considering regional depositional patterns and groundwater occurrence.
- Acquisition of resistivity data using multi-electrode systems.
- Preparation of topographically corrected inverse resistivity sections.
- Interpretation of subsurface geological and hydrogeological conditions.
- Delineation of aquifer boundaries and groundwater-bearing formations.
- Recommendations for borehole siting and drilling depth.



Part -D: Hydrogeological Interpretation and Recommendations

- Integrate geological, hydrogeological, VES and ERT findings.
- Identify groundwater potential zones.
- Recommend borehole locations.
- Recommend drilling methodology and target depth.
- Assess groundwater development risks.
- Provide groundwater resource management and monitoring recommendations.

Scope of the assignment

Under this Long-Term Agreement (LTA), the specific scope of work for each assignment shall be defined through individual work orders or assignment-specific Terms of Reference (ToR), based on programmatic priorities and emerging needs. The consulting agency shall provide specialized technical services related to hydrogeological investigation services, including Vertical Electrical Sounding (VES), Electrical Resistivity Tomography (ERT), groundwater assessments, borehole siting investigations, aquifer characterization, and technical advisory services on an as-needed basis during the agreement period.

The scope of work shall include, but not be limited to, the following:

A. Geological and Hydrogeological Investigations

- Geological assessments.
- Aquifer characterization.
- Review of groundwater data.
- Borehole siting studies.

B. Geophysical Surveys

- VES surveys.
- ERT surveys.
- Data acquisition and interpretation.
- Groundwater potential mapping.

C. Groundwater Development Support

- Borehole location selection.
- Drilling depth recommendations.
- Borehole design support.
- Technical review during drilling activities.

D. Capacity Building and Technical Advice

The consulting agency may also be requested to provide technical support services under assignment-specific work orders, including but not limited to:

- Support NWSC and municipalities.
- Technical reviews.
- Hydrogeological advisory services.
- Training and knowledge sharing.

Geographic Coverage

Assignments under this Long-Term Agreement may be undertaken in municipalities and rural municipalities where WaterAid Nepal and its implementing partners are currently working, have previously worked, or may work in the future. Activities are expected to be primarily focused on selected Nepal Water Supply Corporation (NWSC) branch offices operating in Madhesh Province, as well as Water Supply and Sanitation User Committee (WSSUC) managed water supply systems in Madhesh Province. Assignments may also be undertaken in Godawari Municipality of Bagmati Province and other locations identified by WaterAid Nepal based on programmatic priorities and emerging needs

Deliverables

The deliverables described below are indicative and may vary depending on the scope, geographic coverage, and requirements of individual work orders issued under this Long-Term Agreement.

I. Geological & Hydrogeological Package

- Geological review report.
- Existing groundwater data inventory.
- Hydrogeological assessment report.

II. VES Survey Package

- Raw survey data.
- VES curves.
- Interpretation report.
- Recommended drilling depths.

III. ERT Survey Package

- Processed resistivity profiles.
- Inverse resistivity sections.
- Geological interpretation.
- Aquifer delineation maps.
- Recommended drilling depths.

IV. Groundwater Development Package

- Recommended borehole locations.
- Borehole design inputs.
- Drilling methodology recommendations.

V. Final Technical Report

- Methodology.
- Findings.
- Maps and figures.
- Conclusions.
- Recommendations.

Timeline / Period of Service

The LTA shall remain valid for a period of three (3) years from the date of signing, unless terminated earlier in accordance with the agreement provisions which depends on satisfactory performance, consistent compliance to the minimum criteria as stipulated under the expected competencies, availability of funding and annual performance review by WaterAid Nepal.

During the validity of the LTA, WaterAid Nepal will issue assignment-specific work orders or ToR based on programmatic needs. The duration of each assignment will depend on its specific scope of work and will be specified in the respective ToR issued under the LTA. The Consulting agency shall complete all activities within the timelines agreed for each assignment.

Expected Competencies

The Consulting agency is expected to have following competencies -

- Minimum **5 years of proven experience** in hydrogeological investigations, groundwater assessment, borehole siting, and geophysical surveys (VES and/or ERT).
- The proposed team shall comprise **Team Leader/Senior Hydrogeologist, Geophysicist, Hydrogeologist, Field Survey Engineers/Technicians, and Data Processing & GIS Specialist.**
- Team Leader/Senior Hydrogeologist shall possess a **Master's degree or higher in Hydrogeology, Geology, Geophysics, Water Resources Engineering, or related discipline** with a minimum of **7 years of relevant professional experience.**
- Geophysicist shall possess a **Bachelor's degree or higher in Geophysics, Geology, Hydrogeology, Engineering, or related discipline** with a minimum of **5 years of experience in VES and ERT surveys and interpretation.**
- Hydrogeologist shall possess a **Bachelor's degree or higher in Hydrogeology, Geology, Water Resources Engineering, or related discipline** with a minimum of **5 years of experience in groundwater assessment, aquifer characterization, and borehole design.**
- Field Survey Engineers/Technicians shall possess a **Diploma or Bachelor's degree in Engineering, Geology, Geophysics, or related discipline** with demonstrated experience in geophysical field investigations and operation of resistivity survey equipment.
- Data Processing & GIS Specialist shall possess a **Bachelor's degree or higher in GIS, Geomatics, Engineering, Geology, or related discipline** with experience in

survey data processing, spatial analysis, and preparation of technical maps and figures.

- The consulting agency shall have successfully completed **at least five similar assignments** related to groundwater exploration, hydrogeological investigation, VES, ERT, or borehole siting studies.
- The consulting agency shall demonstrate **excellent analytical, interpretation, reporting, and technical documentation skills** relevant to groundwater investigations.
- The consulting agency shall possess or have access to the **necessary geophysical equipment, software, and technical resources** required to undertake VES and ERT surveys and data interpretation.
- The consulting agency shall submit copies of **company registration, PAN/VAT registration, latest tax clearance certificate, and any applicable professional licenses or permits**.
- The consulting agency shall maintain an **Occupational Health and Safety (OHS) policy/system** and demonstrate commitment to safe field operations.

Documents for Submission

- Technical and financial Proposal outlining methodology, detailed budget and CV of proposed resource persons
- Organisation profile
- Company registration, VAT/PAN certificate, and latest year tax clearance(2081/82)/ tax exemption certificates, valid operating licenses.
- Documents verifying the expected competencies as stipulated in above section
- Relevant experience in facilitating/supporting similar hydrogeological investigation services, including Vertical Electrical Sounding (VES), Electrical Resistivity Tomography (ERT), groundwater assessments, borehole siting investigations, aquifer characterization, and technical advisory services.
- Self-declaration by the applicant company/firm in the format attached to **Annex A** in this TOR.
- Financial proposal as per attached Financial Proposal format **Annex B** (NPR, including all taxes) - Unit rate for Vertical Electrical Sounding (VES), Electrical Resistivity Tomography (ERT), and technical advisory services

Reporting

Reporting requirements may vary depending on the individual assignment and will be detailed in each assignment-specific work order or ToR issued under the LTA. Reporting requirements may include, but are not limited to:

- **Inception Report:** Methodology, staffing, implementation schedule, survey design, quality assurance approach, and work plan.
- **Mobilization Report:** Confirmation of site selection, stakeholder coordination, equipment deployment, and field readiness.

- **Progress Reports:** Progress against agreed milestones, completed activities, preliminary findings, challenges encountered, mitigation measures, and planned next steps.
- **Geological and Hydrogeological Investigation Report:** Review of existing geological information, borehole records, hydrogeological conditions, and preliminary assessment of groundwater potential
- **VES Survey Report:** Survey methodology, field data, interpreted VES curves, resistivity analysis, lithological interpretation, aquifer characterization, and recommended drilling depths.
- **ERT Survey Report:** Survey methodology, processed resistivity data, inverse resistivity sections, hydrogeological interpretation, aquifer delineation, and groundwater potential assessment.
- **Groundwater Assessment Report:** Integrated interpretation of geological and geophysical findings, recommended borehole locations, target drilling depths, anticipated yield, and source development recommendations.
- **Completion and Handover Report:** Summary of activities undertaken, findings, recommendations, and formal handover of all datasets, maps, models, drawings, and technical documentation.
- All reports shall be submitted in editable electronic formats (Word, Excel, GIS/CAD where applicable), together with PDF versions.

Proposal Preparation and Submission

Consulting agencies are requested to submit their Technical and Financial Proposals electronically to WaterAid Nepal in accordance with the following instructions:

- The Technical and Financial Proposal shall be submitted electronically to nepal-procurement@wateraid.org with the subject line: "**Proposal Submission for Long-Term Agreement (LTA) for Hydrogeological Investigation, Groundwater Assessment, Vertical Electrical Sounding (VES), Electrical Resistivity Tomography (ERT), Borehole Siting and Groundwater Development Services.**"
- Proposals submitted to any email address other than the one specified above shall not be considered.
- Competing consulting agencies shall uphold the highest standards of integrity, transparency, and professional ethics throughout the procurement process.
- Any attempt to influence the proposal evaluation or selection process shall result in disqualification.
- Any misrepresentation of qualifications, experience, staff resources, equipment availability, or legal status may result in rejection of the proposal.
- The Technical Proposal shall include:
 - ✓ Proposed methodology and approach.
 - ✓ Understanding of the assignment.
 - ✓ Team composition and CVs.
 - ✓ Relevant project experience.

- ✓ Available survey equipment and software.
- ✓ Quality assurance and Occupational Health and Safety (OHS) measures
- The Financial Proposal shall follow the format provided in Annex B and clearly indicate all applicable fees, taxes, VAT, and reimbursable costs.
- The proposal shall be accompanied by:
 - ✓ Company/Firm Registration Certificate.
 - ✓ PAN/VAT Registration Certificate.
 - ✓ Latest Tax Clearance Certificate.
 - ✓ Valid operating licenses (if any).
 - ✓ Evidence of similar assignments completed.
 - ✓ Self-declaration form (Annex A).

Confidentiality and Data Ownership

All information, analysis, reports, datasets, and outputs produced under this agreement shall remain the property of WaterAid Nepal.

The service provider shall:

- Maintain confidentiality of all information.
- Not to disclose results to third parties without written approval from WAN.
- Secure all records and electronic datasets.

Ethical Standards and Conflict of Interest

The service provider shall:

- Avoid actual or perceived conflicts of interest.
- Comply with applicable laws and professional standards.
- Adhere to WaterAid safeguarding, anti-fraud, anti-corruption policies, Anti-money laundering policies.
- Adhere to the principles of human rights, modern slavery, gender equality, Prevention of sexual exploitation, abuse and harassment (PSEAH).

Payment Terms

Payments shall be made based on the specific scope, deliverables, and level of work defined under each assignment specific work order or ToR issued by WAN. Payments will be processed upon satisfactory completion and approval of agreed deliverables, in accordance with WAN procedures and after submission of required supporting documents and invoices by the Consulting agency.

Safeguarding

The selected Consulting agency must adhere to the WaterAid's global code of conduct and children safeguarding requirements as Safeguarding standard and policies at WaterAid which is found here www.wateraid.org/uk/safeguarding. All personnel deployed under this LTA shall complete WaterAid's safeguarding orientation prior to field deployment where required.

Contact person

Er. Dharma Ratna Chitrakar, Senior Technical Manager from WaterAid Nepal will be available to answer any queries or clarifications you might have regarding the deliverables at DharmaRatnaChitrakar@wateraid.org

To Apply

A pre-proposal meeting may be conducted, if required, to provide clarification on the assignment. The proposal submission date is at the latest by **14 September 2026**. Any clarity on the term of reference can be inquired through written email through following email. Therefore, interested agencies are requested to submit their proposal through email nepal-procurement@wateraid.org.

Proposals must be submitted in PDF format via email to nepal-procurement@wateraid.org with the subject line: **"Proposal Submission for Long-Term Agreement (LTA) for Hydrogeological Investigation, Groundwater Assessment, Vertical Electrical Sounding (VES), Electrical Resistivity Tomography (ERT), Borehole Siting and Groundwater Development Services."**

However, for each assignment-specific work order or ToR, consultant / service provider will be invited to submit separate Technical and Financial Proposals. The proposal submission deadline and other instructions will be specified in the respective work order or ToR.

ANNEXES

Annex A

Form for Submitting Self- Declaration

(This form must be submitted only using the contractor's official letterhead)

WaterAid Nepal (WAN)

.....

.....

Dear Sir/Madam,

We hereby declare that our organization has the operational capacity and the operational resources (technical, management) to successfully fulfil the requirement in accordance with ToR and:

- We do not have any personal or financial interest with any staff member at WAN.
- We hereby declare that all the information that comes to my possession and that is deliberated upon during the procurement process, especially during the evaluation, shall not be disclosed to any other party.
- We have never ever been blacklisted and/ or there was no debarring action against us for any default in supply of materials/ equipment.
- We will maintain strict confidentiality of all financial and organizational information and adhere to WaterAid's ethical standards and data protection policies."
- We have not enforced or exploited child labour or and forced labour.

We confirm that the declaration I have made above is, to the best of my knowledge are correct. I fully understand that, if WAN concludes that the declarations, I have made are false or materially misleading, WAN may refer the matter to the relevant legal authorities for them to investigate and to take appropriate legal action.

Name:

Title or position in the applicant organization:

Signature and official stamp of applicant:

Date

Annex B

Financial Proposal Submission Form

Name of Consulting agency.: _____

Address: _____

Contact Person: _____

Telephone/Email: _____

Validity of Rates: _____ Months/Years

The financial proposal should be based on **unit rates** rather than a lump-sum budget because WaterAid Nepal will issue assignment-specific work orders over the contract period.

TABLE B-1 [For Madhesh Province]

SN.	Description	Unit	Unit Rate (NPR) (With VAT) – FY ¹ 2026/27	Unit Rate (NPR) (With VAT) – FY 2027/28	Unit Rate (NPR) (With VAT) – FY 2028/29
1	Vertical Electrical Sounding (VES) Survey, Data Processing, Interpretation with recommended borehole siting and target Depth assessment	Per Point			
2	Electrical Resistivity Tomography (ERT) Survey up to 1 km profile length including interpretation with recommended borehole siting and target Depth assessment	Per Profile			

¹ FY = Fiscal Year of WaterAid Nepal, running from 1 April 2026 to 31 March 2027 for FY 2026/27, and correspondingly from 1 April to 31 March for subsequent fiscal years.

3	Additional ERT Survey Length beyond 1 km	Per 100 m			
4	Groundwater Assessment and Aquifer Characterization	Per Assignment			
5	Borehole Drilling Supervision and Technical Verification	Person-Day			
6	Team Leader / Senior Hydrogeologist	Person-Day			
7	Geophysicist	Person-Day			
8	Hydrogeologist	Person-Day			
9	GIS and Data Processing Specialist	Person-Day			
10	Survey Technicians	Person-Day			

TABLE B-2 [For Godawari Municipality, Bagmati Province]

SN.	Description	Unit	Unit Rate (NPR) (With VAT) – FY ² 2026/27	Unit Rate (NPR) (With VAT) – FY 2027/28	Unit Rate (NPR) (With VAT) – FY 2028/29
1	Vertical Electrical Sounding (VES) Survey, Data Processing, Interpretation with recommended borehole siting and target Depth assessment	Per Point			
2	Electrical Resistivity Tomography (ERT) Survey up to 1 km profile length including interpretation with	Per Profile			

² FY = Fiscal Year of WaterAid Nepal, running from 1 April 2026 to 31 March 2027 for FY 2026/27, and correspondingly from 1 April to 31 March for subsequent fiscal years.

	recommended borehole siting and target Depth assessment				
3	Additional ERT Survey Length beyond 1 km	Per 100 m			
4	Groundwater Assessment and Aquifer Characterization	Per Assignment			
5	Borehole Drilling Supervision and Technical Verification	Person-Day			
6	Team Leader / Senior Hydrogeologist	Person-Day			
7	Geophysicist	Person-Day			
8	Hydrogeologist	Person-Day			
9	GIS and Data Processing Specialist	Person-Day			
10	Survey Technicians	Person-Day			

Financial Proposal Declaration

We hereby certify that the rates quoted above are inclusive of all costs necessary for the satisfactory delivery of services under this Long-Term Agreement, including VAT. The proposed rates shall remain valid for the duration specified in the proposal and shall be used as the basis for assignment-specific work orders issued by WaterAid Nepal.

Authorized Signatory: _____

Name: _____

Position: _____

Date: _____

Official Stamp and Signature: _____